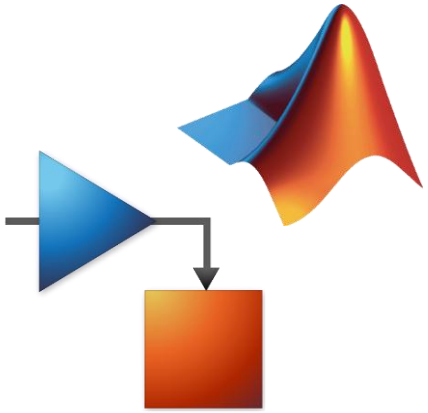
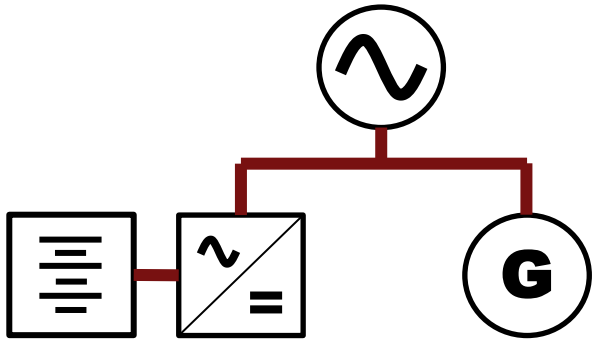
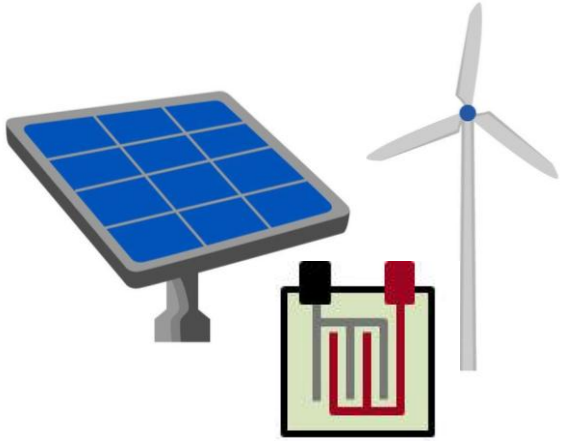
Microgrid Energy Management System Development Using Optimization-based Methods









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MathWorks, Inc.

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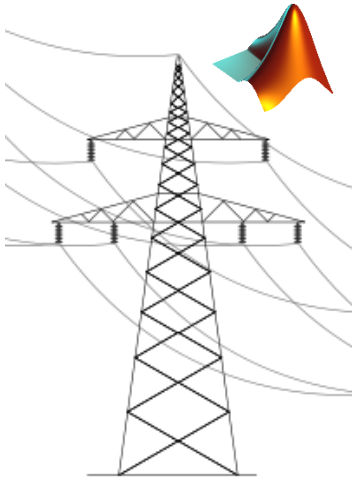
Demo – Microgrid EMS

Using Model-Based Design and Optimization

HEMS example – *microgrid with loads*

PV Panels:
- MPPT Control
- Power electronics

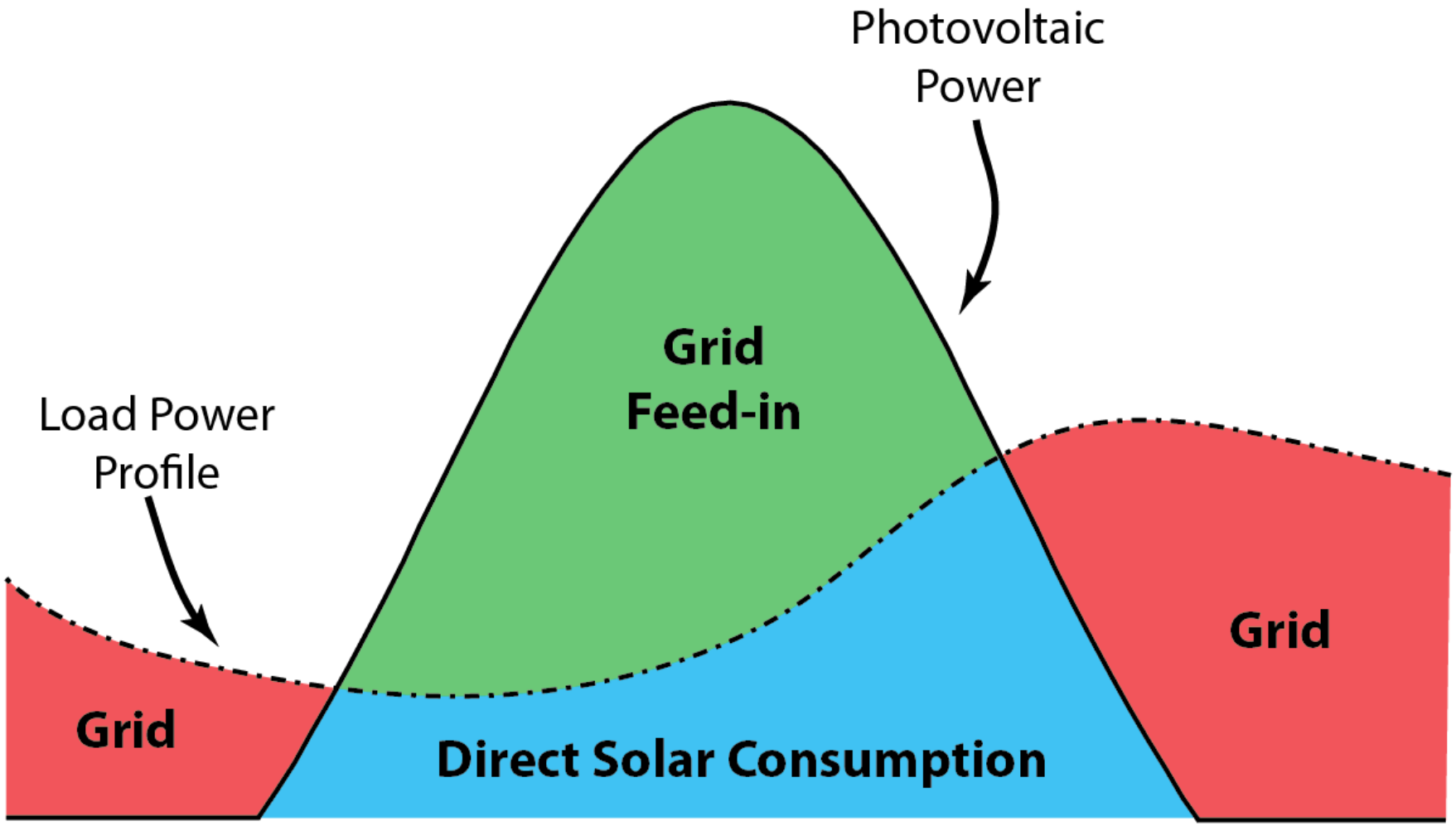




Decision Logic:
- Store/draw power
- Use external grid
- Optimize energy

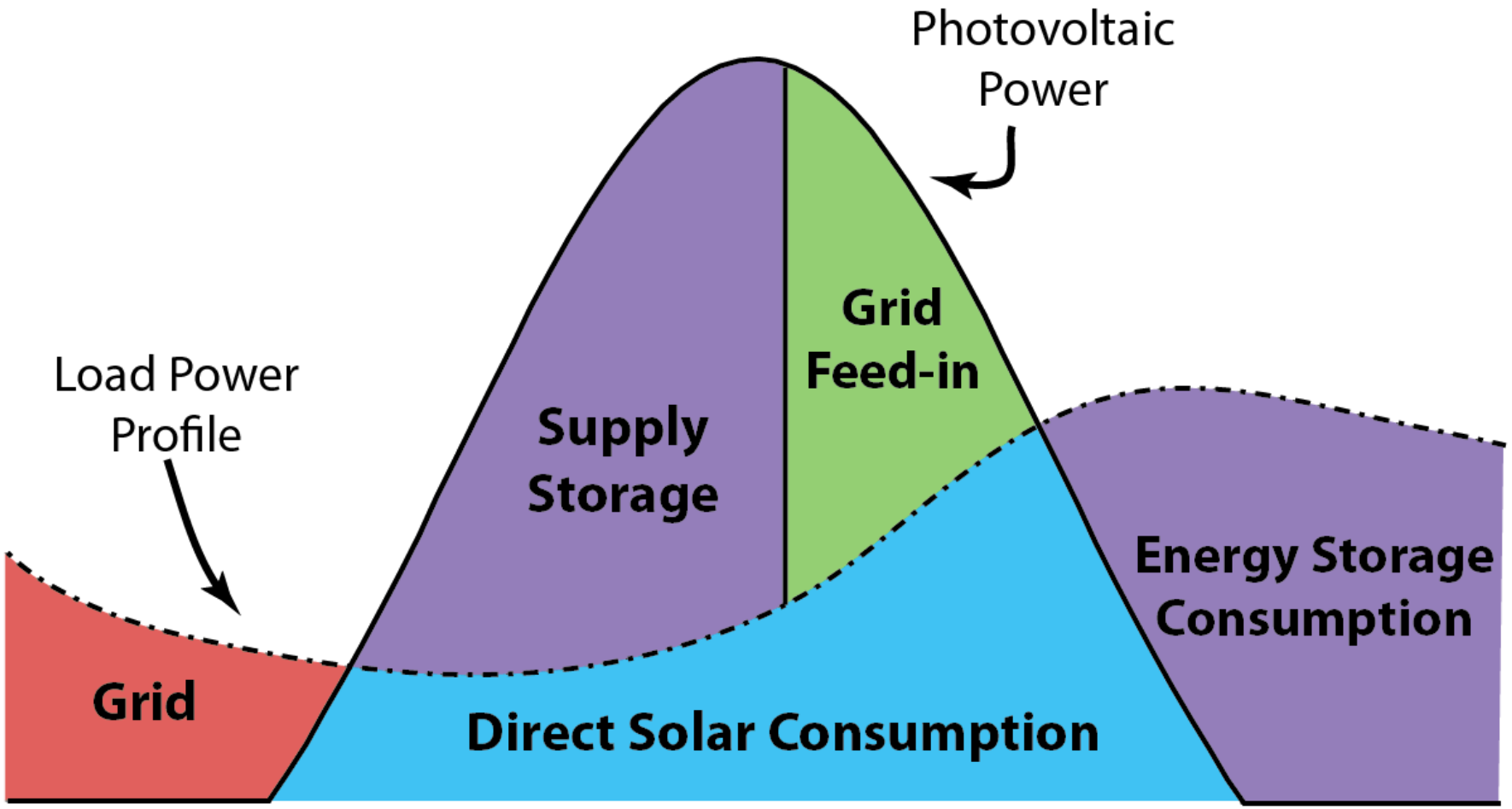
Battery System:
- Charge controls
- Discharge controls
- Power electronics

Why energy storage with solar?



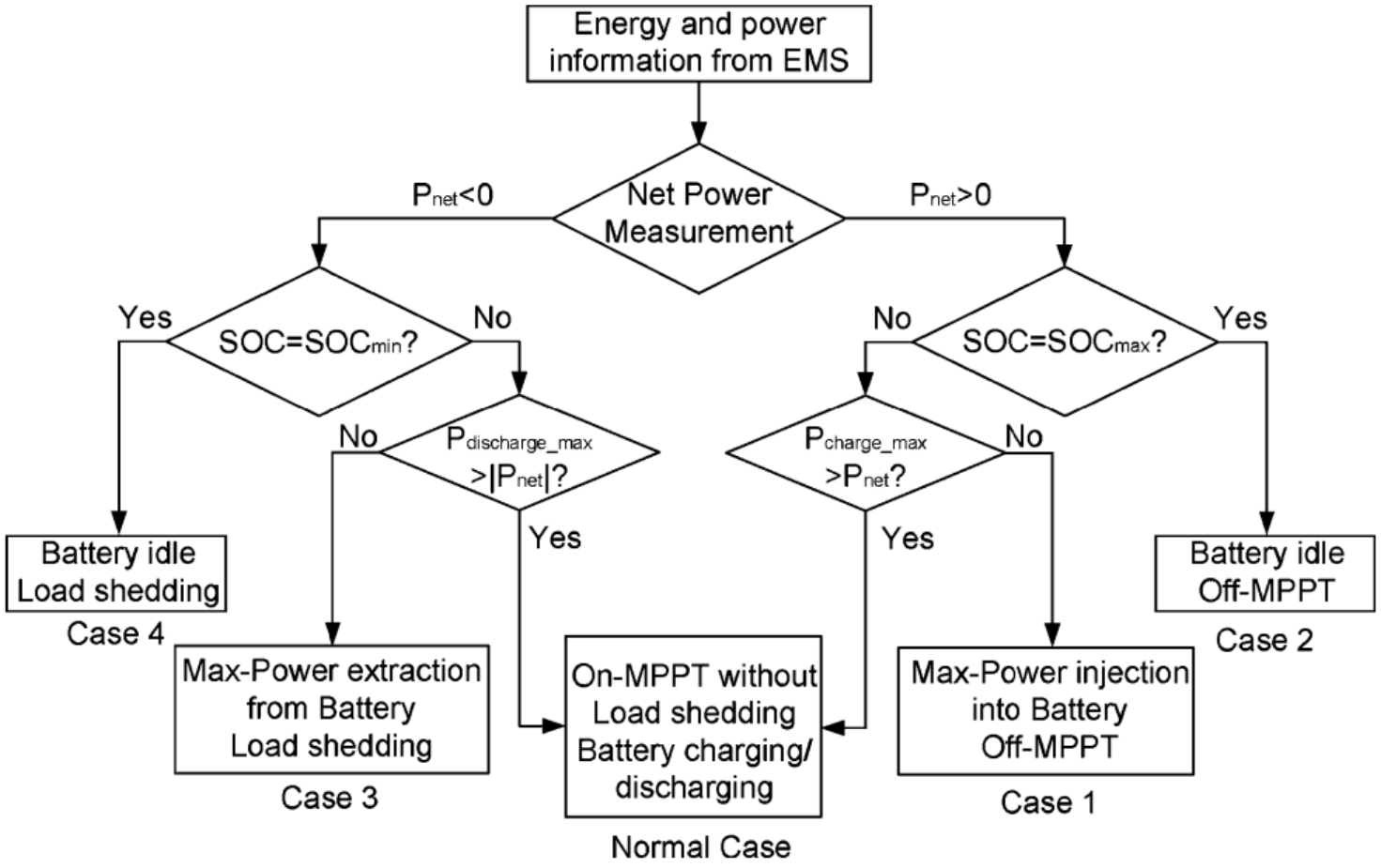
Adapted from: Smart EnergySystems Website
http://www.smart-energy.ag/products/ac-gekoppelte-speicherlosung-smartenergy-ac/?lang=en

Peak Demand Shift using Energy Storage



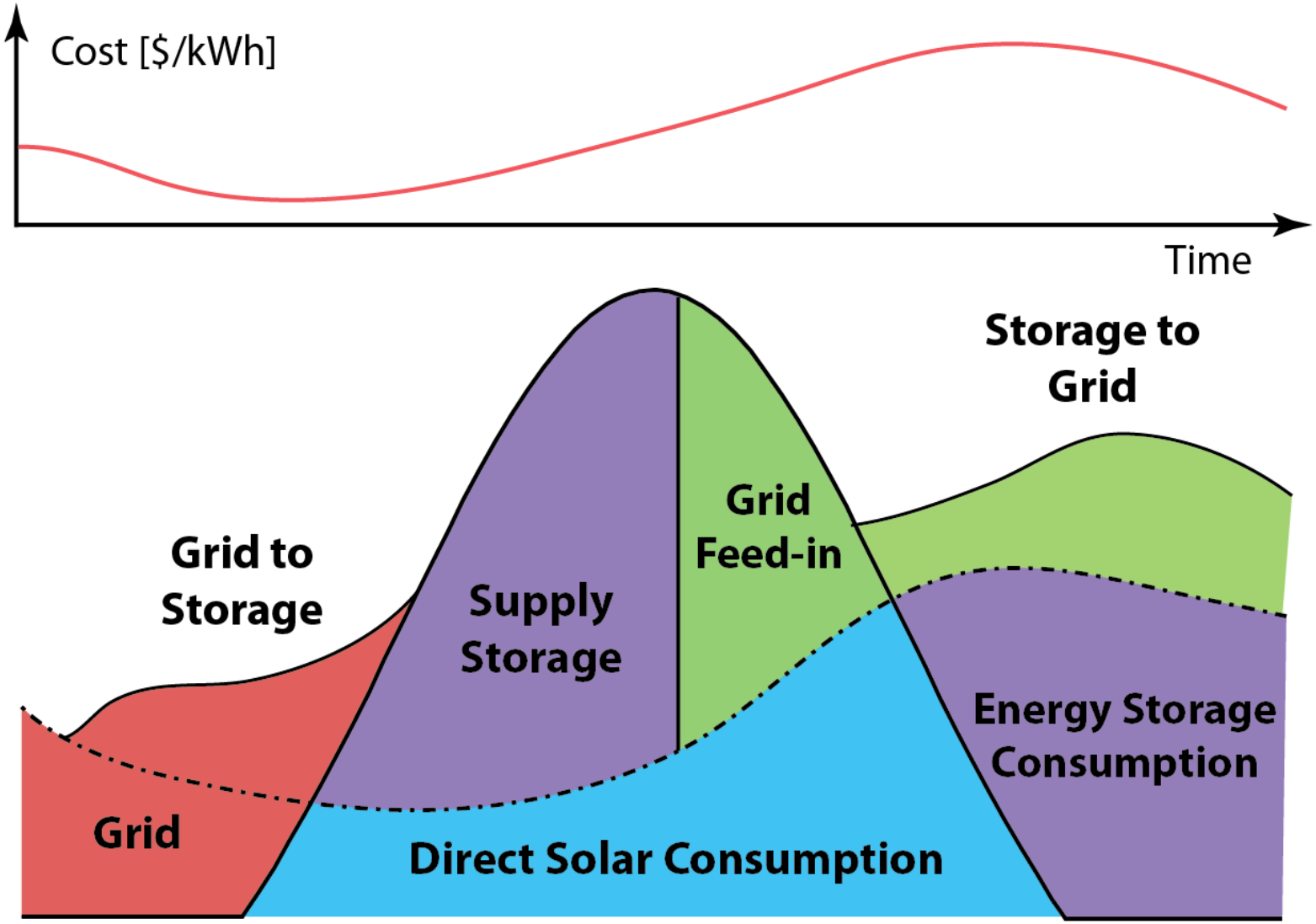
Adapted from: Smart EnergySystems Website
http://www.smart-energy.ag/products/ac-gekoppelte-speicherlosung-smartenergy-ac/?lang=en

Implementation of Energy Management Logic



Ref: Liu 2011 - A Hybrid AC/DC Microgrid and Its Coordination Control

Factoring in Variable Electricity Cost



Adapted from: Smart EnergySystems Website
http://www.smart-energy.ag/products/ac-gekoppelte-speicherlosung-smartenergy-ac/?lang=en

Optimization-based Peak Demand Shift

- Objective function: *what do we want to minimize/optimize?*
  - Minimize total cost of variable priced electricity

$$C_{tot} = \sum_{k=0}^{N} C_{grid}(k) \cdot E_{grid}(k)$$

- Constraints: *how does our system operate?*
  - Power input/output to battery

$$E_{batt}(k) = E_{batt}(k-1) + P_{batt}(k)\Delta T$$

  - Power balance

$$\underbrace{P_{pv}(k) + P_{grid}(k) + P_{batt}(k)}_{\text{Sources}} = \underbrace{P_{load}(k)}_{\text{Loads}}$$

Linear program-based Optimization

- Standard form for the Linear program (LP) - linprog

$$\min_x f^T x \quad \text{such that} \quad \begin{cases} A \cdot x \leq b \\ A_{eq} \cdot x = b_{eq} \end{cases}$$

- Define states (x) necessary for LP optimization
  - P_{grid}(1:N) – Power from grid used from time step 1 to N
  - P_{batt}(1:N) – Power from battery
  - E_{batt}(1:N) – Energy stored in battery

$$x = [P_{grid}(1:N) \quad P_{batt}(1:N) \quad E_{batt}(1:N)]^T$$

Linear program-based Optimization

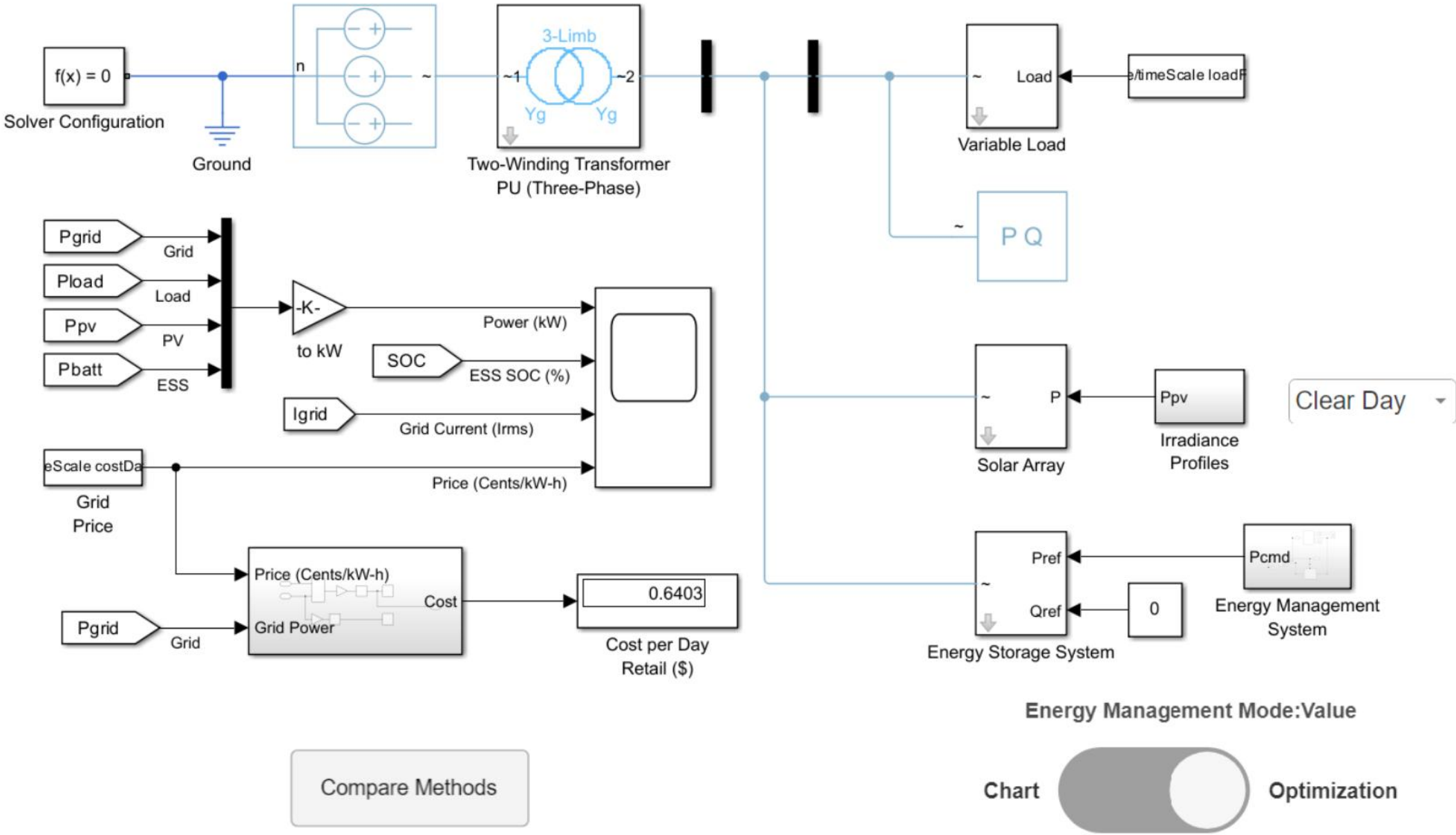
- Equivalent constraint

$$\underbrace{\begin{bmatrix} I_{N\times N} & I_{N\times N} & 0_{N\times N} \\ O_{N\times N} & \Upsilon_{N\times N} & \Phi_{N\times N} \end{bmatrix}}_{A_{eq}} \mathrm{x} = \underbrace{\begin{bmatrix} P_{load}(1:N) - P_{pv}(1:N) \\ E_{batt}(1) \\ 0_{N-1} \end{bmatrix}}_{b_{eq}}$$

$$\Upsilon_{3\times 3} = \begin{bmatrix} 0 & 0 & 0 \\ \Delta T & 0 & 0 \\ 0 & \Delta T & 0 \end{bmatrix} \qquad \Phi_{3\times 3} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

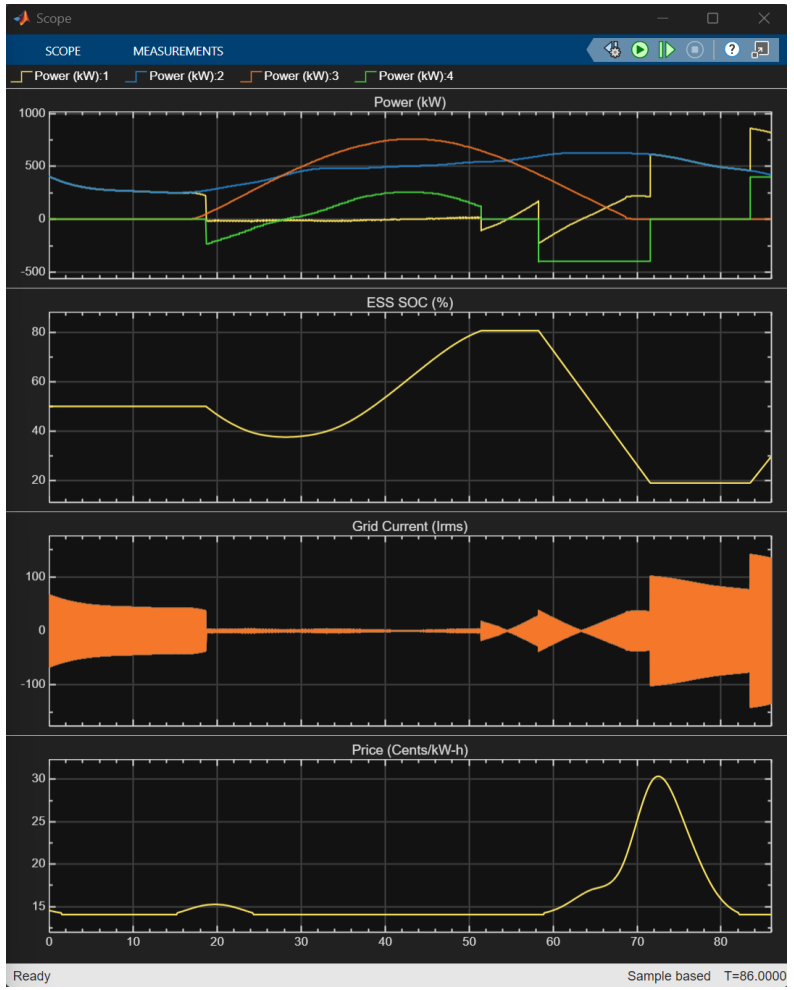
- Inequality constraints

$$\underbrace{\begin{bmatrix} O_{N\times N} & I_{N\times N} & O_{N\times N} \\ O_{N\times N} & -I_{N\times N} & O_{N\times N} \\ O_{N\times N} & O_{N\times N} & I_{N\times N} \\ O_{N\times N} & O_{N\times N} & -I_{N\times N} \end{bmatrix}}_{A} \mathrm{x} \geq \underbrace{\begin{bmatrix} P_{max} \\ -P_{min} \\ E_{max} \\ -E_{min} \end{bmatrix}}_{b}$$


f(x) = 0
Solver Configuration
Ground
n
~
3-Limb
~1
~2
Yg
Yg
Two-Winding Transformer
PU (Three-Phase)
~
Load
Variable Load
timeScale loadP
~
P Q
Pgrid
Grid
Pload
Load
Ppv
PV
Pbatt
ESS
-K-
to kW
Power (kW)
SOC
ESS SOC (%)
Igrid
Grid Current (Irms)
eScale costDa
Price (Cents/kW-h)
Grid
Price
~
P
Ppv
Solar Array
Irradiance
Profiles
Clear Day
Price (Cents/kW-h)
Cost
Pgrid
Grid
Grid Power
0.6403
Cost per Day
Retail ($)
Pref
Qref
~
0
Energy Storage System
Pcmd
Energy Management
System
Energy Management Mode:Value
Chart
Optimization
Compare Methods

Policy Comparison

Heuristic



Optimized

